

Load-Aware A* Path Planning for Quadruped Robots

Day 22 of 30 · · Week 4: Results Review & Integration · Member 1 — Results Section Input

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1. Nature of Work Done

Day 22 initiates Week 4 activity for Member 1. With Methodology v2.0 locked and all Week 3 deliverables handed off to Member 4 (Day 21), the focus shifts to reviewing the results being generated by Member 4, providing interpretive commentary grounded in the algorithm design, and cross-checking the accuracy of results against the theoretical model. No new algorithmic work is introduced. The four tasks below are diagnostic and advisory in nature.

Task	Description	Status
Task 1	Review Results Section Progress	In Progress
Task 2	Provide Inputs on Load-Aware A* Findings	In Progress
Task 3	Help Interpret Results (20 Scenarios)	In Progress
Task 4	Check Accuracy Against Methodology v2.0	In Progress

2. Results Section Progress Review

Based on the 20-scenario test matrix submitted on Day 19 and the metric definitions established in Methodology v2.0 (Section III-F), the following review framework is applied to Section V (Results) as it is being populated by Member 4.

2.1 Expected Structure of Section V

Section V should present the 20-scenario results table and summary statistics in the following sub-structure:

- V-A: 20-Scenario Raw Results Table — all 9 metrics (L, C, N, T, S_avg, M_min, D, dL, dT%) for both Standard A* and Load-Aware A* side by side
- V-B: Aggregate Statistics — mean, min, max, and standard deviation for primary metrics (S_avg, M_min, dL, dT%) across all 20 scenarios
- V-C: Statistical Significance — paired t-test results for S_avg and M_min comparing Standard A* vs Load-Aware A* (p-value, confidence interval)
- V-D: Path Divergence Analysis — count of scenarios where D = True vs False, correlated with payload mass (mp = 3, 5, 7, 9 kg)
- V-E: Visualisation — representative path diagrams for at least 2 contrasting scenarios (one with D = True, one with D = False)

2.2 Completeness Checklist — Member 4 Handoff

Item	Required From Day 19 Matrix	Expected in Section V	Verified
Raw metric table	20 rows × 9 columns per algorithm	Table V-A	Pending Member 4
S_avg comparison	Lower on Load-Aware paths	Section V-B / V-C	Pending Member 4
M_min comparison	> 0.05 m on Load-Aware vs ≤ 0.05 m on Standard	Section V-B / V-C	Pending Member 4
D flag correlation	D = True at mp ≥ 5 kg	Section V-D	Pending Member 4
dL range	Expected 1–3 steps extra	Section V-B	Pending Member 4
dT% range	Expected < 5%	Section V-B	Pending Member 4

3. Inputs on Load-Aware A* Findings

The following interpretive inputs are provided to Member 4 to guide accurate write-up of the Section IV and VI findings. These are derived directly from the algorithmic analysis completed in Days 18–20.

3.1 Primary Finding: Stability-Cost Trade-off

The central finding of Load-Aware A* is the deliberate trade-off between geometric efficiency and mechanical safety. Member 4 should frame this as:

- Load-Aware A* accepts marginally longer paths (dL = 1–3 steps) in exchange for substantially improved stability metrics
- The stability-cost trade-off is quantified by: ΔS_{avg} (reduction in mean stability penalty) and ΔM_{min} (improvement in minimum margin)
- When dL = 0 (same path length), Load-Aware A* has found an equally short path that is also more stable — this is the strongest result scenario
- When dL ≥ 3, the detour cost should be evaluated against the M_min improvement to confirm the trade-off remains justified

3.2 Alpha Parameter Interpretation

The empirically selected weight $\alpha = 1.0$ has specific implications for how results should be reported:

α Value	Algorithmic Behaviour	Implication for Results
$\alpha = 0$	Identical to Standard A*	Baseline — dL = 0, D = False by definition
$\alpha = 0.5$	Weak stability bias	Divergence rare; S_avg improvement marginal
$\alpha = 1.0$ (selected)	Balanced stability-cost weight	Consistent divergence at mp ≥ 5 kg; justified dL
$\alpha = 1.5\text{--}2.0$	Strong stability bias	Risk of unnecessarily long paths; dL may be unjustified

3.3 Admissibility and Optimality Clarification

A common reporting error is to describe Load-Aware A* as ‘finding a longer path’ without clarifying the cost function. Member 4 must note:

- Load-Aware A* is optimal with respect to $f(n) = g(n) + \alpha \cdot S(n) + h(n)$ — it returns the minimum-cost path under this combined objective
- Standard A* is optimal with respect to geometric length only — it cannot be compared directly on stability
- $dL > 0$ does NOT mean Load-Aware A* is suboptimal — it means the stability-cost objective found a different optimal solution
- The admissibility proof (Day 20) confirms $h(n)$ remains a valid lower bound because $S(n)$ is accumulated only in $g(n)$

4. Results Interpretation Guide — 20-Scenario Analysis

4.1 Reading the Metric Table

When reviewing each scenario row, the following logic chain should be applied:

Step	Check	Expected Pattern	If Unexpected
1	Is $D = \text{True}$?	True for $mp \geq 5$ kg in most scenarios	Check if obstacle density is very low (may not create routing conflict)
2	Is $S_{\text{avg}}(\text{LA}) < S_{\text{avg}}(\text{Std})$?	Yes — primary effectiveness criterion	Flag scenario for individual analysis
3	Is $M_{\text{min}}(\text{LA}) > 0.05$ m?	Yes; $M_{\text{min}}(\text{Std})$ may be ≤ 0.05 m	Verify payload position is in rows 3–7
4	Is $dL \leq 3$?	Yes for $\alpha = 1.0$ on 10×10 grid	Large dL suggests high-penalty zone blocking efficient routes
5	Is $dT\% < 5\%$?	Yes for S_{map} $O(1)$ lookup	Large $dT\%$ may indicate tie-breaking overhead — investigate N

4.2 Scenario Categories

The 20 scenarios naturally cluster into four interpretive categories:

Category	Characteristics	Typical Scenario Count	Key Metric to Report
Strong Improvement	$D = \text{True}$, large ΔS_{avg} , $M_{\text{min}}(\text{LA}) > 0.05$ m, small dL	~8–10 scenarios	Both S_{avg} and M_{min} improvement with $dL \leq 2$
Path-Preserving Safety	$D = \text{False}$ (same path), but $M_{\text{min}}(\text{LA}) \geq M_{\text{min}}(\text{Std})$	~3–4 scenarios	M_{min} improvement with $dL = 0$ — strongest result
High-Detour Justified	$D = \text{True}$, $dL = 3$ – 4 , but large ΔM_{min}	~4–5 scenarios	Show trade-off is justified by safety gain
Low-Payload Baseline	$mp = 3$ kg, $D = \text{False}$ or marginal	~3–4 scenarios	Validates $\alpha = 1.0$ threshold behaviour

4.3 Statistical Interpretation Guidance

For the paired t-test comparing S_{avg} and M_{min} across 20 scenarios:

- H_0 : No difference in mean S_{avg} between Standard A* and Load-Aware A*
- H_1 : Load-Aware A* achieves lower mean S_{avg} (one-tailed)
- Expected: $p < 0.05$ confirming statistical significance at 95% confidence
- For M_{min} : report the proportion of scenarios where $M_{min}(LA) > 0.05$ m vs $M_{min}(Std) \leq 0.05$ m as a binary safety improvement rate
- 20 scenarios is sufficient for t-test validity (Central Limit Theorem applies; normal distribution assumed)

5. Accuracy Check — Methodology v2.0 Compliance

All results reported in Section V must be traceable to Methodology v2.0 (Day 17). The following accuracy checks are issued for Member 4's use during write-up:

5.1 Equation Accuracy Checks

Equation	Formula	Common Error to Avoid
Eq. 4 (CoM)	$x_{CoM} = (m^b \cdot x^b + m^{\square} \cdot x_{payload}) / (m^b + m^{\square})$	Do not use simple midpoint — weights must be mass-proportional
Eq. 5 (S(n))	$S(n) = \ x_{CoM}(n) - x^c(n)\ _2$	Must use Euclidean (L2) norm, not Manhattan distance
Eq. 6 (M(n))	$M(n) = \min_i d(x_{CoM}(n), edge_i)$	Must be perpendicular distance to each edge, not vertex distance
Eq. 10 (cost)	$f(n) = g(n) + \alpha \cdot S(n) + h(n)$	$S(n)$ is added only once in $g(n_{next})$ per Eq. 11, not in $h(n)$
Eq. 11 (g-update)	$g(n_{next}) = g(n_{cur}) + 1 + \alpha \cdot S(n_{next})$	Step cost base is always +1 (uniform); stability is additive

5.2 Parameter Accuracy Checks

Parameter	Correct Value	Impact if Wrong
Grid size	$R = C = 10$ (100 cells)	S_{map} size mismatch
Body mass m^b	10.0 kg	CoM calculation offset for all scenarios
Foot half-span h_s	0.4 m	Support polygon incorrect; $M(n)$ invalid
Stability threshold	$M(n) > 0.05$ m	Safety classification incorrect
α (stability weight)	1.0	Cost function non-standard; results non-reproducible
Payload range $m^{\square}$	3–9 kg (step 2 kg)	Incorrect CoM shift magnitudes
Scenarios	20 (10×10 grid, seeds pre-validated)	Statistical validity compromised if fewer

5.3 Notation Consistency Check

Across Sections III, IV, and V, the following notation must remain uniform (per Methodology v2.0, Table I):

- Always $S(n)$ for stability penalty — never $SP(n)$ or $Penalty(n)$
- Always $M(n)$ for stability margin — never $SM(n)$ or $Margin(n)$
- Always α for stability weight — never w , λ , or α spelled out
- Load-Aware A^* (hyphenated, both words capitalised) — never ‘modified A^* ’ or ‘stability A^* ’
- S_{avg} and M_{min} (subscripts) in all tables — must match metric names defined in Section III-F